1. Remove All Occurences

```
#include <stdio.h>


void remove_All_occurences(char *str, char ch)
{
   int i,j;
   for(i=0;str[i];i++)
   {
      if(str[i]==ch)
      {
         for(j=i;str[j];j++)
         {
            str[j]=str[j+1];
         }
         i--;
      }
   }
}
int main()
{
  char str[100];
  char ch;
  printf("Enter strring: ");
  scanf("%[^\n]",str);
  printf("Enter character: ");
  scanf(" %c",&ch);
  remove_All_occurences(str,ch);
  printf("Modified string is %s\n",str);
  return 0;
}
```

2. Move Zeroes to Last

```
#include <stdio.h>
int main()
{
  int size,i,j,digit=0,count=0;
  printf("Enter size: ");
  scanf("%d",&size);
  int arr[size];
  int brr[size];
  for(i=0;i<size;i++)
  {
     scanf("%d",&arr[i]);
  }
   for(i=0;i<size;i++)
  {
     if(arr[i]!=0)
     {
        brr[count++]=arr[i];
     }
  }
   for(i=0;i<size;i++)
  {
     if(arr[i]==0)
     {
        brr[count++]=0;
     }
  }

  for(i=0;i<size;i++)
  {
```

```
      printf("%d ",brr[i]);
   }
   return 0;
}
```

3. Print kth characters:

```
#include <stdio.h>

void print_kth_character(char *str, int k)
{
   if (k <= 0)
   {
      printf("Invalid value of k.\n");
      return;
   }

   printf("Every %dth character from the beginning: ", k);

   for (int i = k - 1; str[i] != '\0'; i = i + k)
   {
      printf("%c", str[i]);
   }
}

int main()
{
   char str[100];
   int k;
```

```
    printf("Enter string: ");
    scanf("%[^\n]", str);

    printf("Enter k value: ");
    scanf("%d", &k);

    print_kth_character(str, k);

    return 0;
}
```

4. FizzBuzz

```
#include <stdio.h>

int main()
{
    int n;
    printf("Enter n: ");
    scanf("%d",&n);
    if(n<0)
    {
        printf("invalid input\n");
        return 0;
    }
    for(int i=1;i<=n;i++)
    {
        if(i%3==0 && i%5==0)
        {
            printf("FizzBuzz ");
```

```
        }
        else if(i%3==0)
        {
            printf("Fizz ");
        }
        else if(i%5==0)
        {
            printf("Buzz  ");
        }
        else
        {
            printf("%d ",i);
        }
    }

    return 0;
}
```

5. Uppercase Version:

```
#include <stdio.h>

void string_to_upper(char *str)
{
   int i;
   for(i=0;str[i];i++)
   {
      if(str[i]>='a' && str[i]<='z')
      {
         str[i]=str[i]-32;
      }
   }
}

int main()
{
  char str[100];
  printf("Enter string: ");
  scanf("%[^\n]",str);

  string_to_upper(str);


  printf("Uppercase version: %s\n",str);
   return 0;
}
```

6. Sum of Two Index numbers:

```
#include <stdio.h>
void sum(int arr[],int size,int target,int*index1,int*index2)
{
   for(int i=0;i<size;i++)
   {
      for(int j=i+1;j<size;j++)
      {
         if(arr[i]+arr[j]==target)
         {
            *index1=i;
            *index2=j;
         }
      }
   }
}
int main()
{
  int size,target,index1=0,index2=0;
  printf("Enter size: ");
  scanf("%d",&size);
  int arr[size];
  for(int i=0;i<size;i++)
  {
     scanf("%d",&arr[i]);
  }
  printf("Enter target: ");
  scanf("%d",&target);
  sum(arr,size,target,&index1,&index2);
   printf("Indices of two numbers to get %d: %d, %d\n",target,index1,index2);
```

```
    return 0;
  }
```

7. Remove Repeated Characters:

```
#include <stdio.h>
void remove_repeated_chars(char *str)
{
  int i,j,count=0,k=0;
  char result[100];
  for(i=0;str[i];i++)
  {
    if(str[i]>='A' && str[i]<='Z')
    {
      str[i]=str[i]+32;
    }
    count=0;
    for(j=0;str[j];j++)
    {
      if(str[i]==str[j])
      {
        count++;
      }
    }
    if(count==1)
    {
      result[k]=str[i];
      k++;
    }
  }
```

```
    result[k]='\0';
    for(k=0;result[k];k++)
    {
        str[k]=result[k];
    }
    str[k]='\0';
}
int main()
{
 char str[100];
 printf("Enter string: ");
 scanf("%[^\n]",str);
 remove_repeated_chars(str);
 printf("Resulted string after removing string is %s\n",str);
    return 0;
}}
```

8. Reverse Vowels:

```
#include <stdio.h>

int is_vowel(char c)
{
    return (c=='a'||c=='A'||c=='e'||c=='E'||c=='i'||c=='I'||
            c=='o'||c=='O'||c=='u'||c=='U');
}

int main()
{
    char str[100];
    printf("Enter string: ");
```

```
    scanf("%[^\n]", str);

    int i = 0, j = 0;

    // Find string length
    while(str[j] != '\0') j++;
    j--; // j at last character

    // Two-pointer vowel reversing
    while(i < j)
    {
        if(!is_vowel(str[i]))
            i++;
        else if(!is_vowel(str[j]))
            j--;
        else
        {
            char temp = str[i];
            str[i] = str[j];
            str[j] = temp;
            i++;
            j--;
        }
    }

    printf("Resulted string is %s\n", str);
    return 0;
}
```

9. Print Unique subsets:

```
#include <stdio.h>
#include <math.h>

int main()
{
   int n;
   printf("Enter size: ");
   scanf("%d",&n);
   int arr[n];
   if(n<1 || n>10)
   {
      printf("invalid input. Number must be between 1 and 10.");
      return 0;
   }
   for(int i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
      if(arr[i]<-10 || arr[i]>10)
      {
         printf("invalid input. Number must be between 1 and 10.");
         return 0;
      }
      for(int j=0;j<i;j++)
      {
         if(arr[i]==arr[j])
         {
            printf("All elements must be unique\n");
            return 0;
         }
```

```
        }
    }

    int total=pow(2,n);
    for(int mask=0;mask<total;mask++)
    {
        printf("[");
        for(int i=0;i<n;i++)
        {
            if(mask&(1<<i))
            {
                printf("%d ",arr[i]);
            }
        }
        printf("]\n");
    }
    return 0;
}
```

10. Bitwise Operations

```
#include <stdio.h>
int main()
{
   int n,num,set=0,clear=0,toggle=0;

   printf("Enter num: ");
   scanf("%d",&num);
   printf("Enter n: ");
   scanf("%d",&n);

   set=num|(1<<n);
   clear=num&~(1<<n);
   toggle=num^(1<<n);
   printf("After setting bit %d: %d\n",n,set);
   printf("After clearing bit %d: %d\n",n,clear);
   printf("After toggling bit %d: %d\n",n,toggle);
   return 0;
}
```

11. Find min and max of 2D array:

```
#include <stdio.h>
int main()
{
   int r1,c1,i,j;
   printf("Enter r1 and c1: ");
   scanf("%d%d",&r1,&c1);
   int arr[r1][c1];
   for(i=0;i<r1;i++)
   {
      for(j=0;j<c1;j++)
      {
         scanf("%d",&arr[i][j]);
      }
   }
   int max=arr[0][0];
   int min=arr[0][0];
   for(i=0;i<r1;i++)
   {
      for(j=0;j<c1;j++)
      {
         if(arr[i][j]>max)
         max=arr[i][j];
         if(arr[i][j]<min)
         min=arr[i][j];
      }
   }
   printf("Minimum Element: %d\n",min);
   printf("Maximum Element: %d\n",max);
   return 0;
```

```
}
```

12. Print N prime Number:

```
#include <stdio.h>
int main()
{
   int n,num=2;
   printf("Enter n: ");
   scanf("%d",&n);
   while(n>0)
   {
      int flag=0;
   for(int i=2;i<num;i++)
   {
      if(num%i==0)
      {
         flag=1;
         break;
      }
   }
   if(flag==0)
   {
      printf("%d ",num);
      n--;
   }
   num++;
}

   return 0;
}
```

13. Insert New Values:

```
#include <stdio.h>
int main()
{
   int n,i,insert=0,j;
   printf("Enter n: ");
   scanf("%d",&n);
   printf("Enter element to insert: ");
   scanf("%d",&insert);
   int arr[n+1];
   for(i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }

   for(i=n;i>0;i--)
   {
      arr[i]=arr[i-1];
   }
   arr[0]=insert;
   n++;

   for(i=0;i<n-1;i++)
   {
      for(j=0;j<n-i-1;j++)
      {
         if(arr[j]>arr[j+1])
         {
            int temp=arr[j];
            arr[j]=arr[j+1];
```

```
            arr[j+1]=temp;
          }
       }
    }
    for(i=0;i<n;i++)
    {
       printf("%d ",arr[i]);
    }

    return 0;
}
```

14. String Weight:

```
#include <stdio.h>
int print_weight(char *str)
{
   int i,weight=0;
   for(i=0;str[i];i++)
   {
      weight=weight+str[i];
   }
   return weight;
}
int main()
{
    char str[100];
    printf("Enter string: ");
    scanf("%[^\n]",str);
    printf("String weight: %d\n",print_weight(str));
    return 0;
}
```

15. Print Indexes of all occurences of Substring:

```
#include <stdio.h>
#include<string.h>
void find_occurences_index(char *text,char*pattern)
{
   int len1=strlen(text);
   int len2=strlen(pattern);
  int i,flag,index=-1,j;
  for(i=0;i<=len1-len2;i++)
  {
     flag=0;
     for(j=0;j<len2;j++)
     {
        if(text[i+j]!=pattern[j])
        {
           flag=1;
           break;
        }
     }
     if(flag==0)
     {
        printf("%d ",i+1);
     }
  }
}
int main()
{
   char text[100];
   char pattern[10];
   printf("Enter string: ");
```

```
    scanf("%[^\n]",text);
    printf("Enter word: ");
    scanf(" %[^\n]",pattern);
    find_occurences_index(text,pattern);
    return 0;
}
```

16. Wap For Seconds Using Macro For years:

```
#include <stdio.h>
#define SECONDS(year) ((year)*(365)*(24)*(60)*(60))
int main()
{
    int years;
    printf("Enter years: ");
    scanf("%d",&years);

    printf("Seconds in the year :%d \n",SECONDS(years));

    return 0;
}
```

17. Print Month and its days:

```
#include <stdio.h>

int main()

{

   int month;

   printf("Enter years: ");

   scanf("%d",&month);


   if(month==1||month==3||month==5||month==7||month==8||month==10||month==12)

   {

      printf("The number of days in the month is %d is 31.",month);

   }

   else if(month==2)

   {

      printf("The number of days in the February month is 28 or 29 (depending on whether its a leap
yaer or not)\n");

   }

   else if(month==4||month == 6|| month==9||month==11)

   {

      printf("The number of days in the month is %d is 30.",month);

   }

   else

   {

      printf("Invalid month number. please enter a number between 1 and 12.");

   }

   return 0;

}
```

18. Pivot Index

```
#include <stdio.h>
#include<stdlib.h>

int main()
{

   int n,total_sum=0,right_sum=0,left_sum=0,i,found=0;
   printf("Enter no of Elements: ");
   scanf("%d",&n);
   int arr[n];
   for(i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }

   for(i=0;i<n;i++)
   {
      total_sum+=arr[i];
   }

   for(i=0;i<n;i++)
   {
      right_sum=total_sum-left_sum-arr[i];
      if(left_sum==right_sum)
      {
         printf("Pivot Index : %d\n",i);
         found=1;
      }
      left_sum=left_sum+arr[i];
```

```
    }
    if(!found)
    {
        printf("Pivot not found\n");
    }

}
```

19. Swap Char and Compare:

```
#include <stdio.h>
#include<stdlib.h>
#include<string.h>
void swap(char *str,int i,int j)
{
      char temp=str[i];
       str[i]=str[j];
       str[j]=temp;
}
int areAlmostEqual(char*str1,char*str2)
{
   int len=strlen(str1);
   int diff1=-1,diff2=-1,i;
   for(i=0;i<len;i++)
   {
       if(str1[i]!=str2[i])
       {
           if(diff1==-1)
           diff1=i;
           else if(diff2==-1)
           diff2=i;
```

```
            else
            return 0;
        }
    }
    if(diff1==-1 && diff2==-1)
    return 1;
    if(diff1!=-1 && diff2!=-1)
    {
    swap(str2,diff1,diff2);
    return strcmp(str1,str2)==0;
    }
    return 0;

}
int main()
{

   char str1[100];
   char str2[100];
   printf("Enter string1: ");
   scanf("%s",str1);
   printf("Enter string 2: ");
   scanf(" %s",str2);
   printf("%s\n", areAlmostEqual(str1,str2)?"true":"false");

}
```

20. Check Mirror Image Not:

```
#include <stdio.h>
#include<stdlib.h>
#include<string.h>

int main()
{

  int n,i,flag=0;
  printf("Enter n: ");
  scanf("%d",&n);

  int arr[n];
  int brr[n];
  for(i=0;i<n;i++)
  scanf("%d",&arr[i]);
    for(i=0;i<n;i++)
  scanf("%d",&brr[i]);

  for(i=0;i<n;i++)
  {
     if(arr[i]!=brr[n-1-i])
     {
        flag=1;
        break;
     }
  }
  if(flag==0)
  {
     printf("Yes\n");
```

```
  }
  else
  {
     printf("No\n");
  }

}
```

21. Difference finding Index

```
#include <stdio.h>
int smallestlargest(int arr[],int n)
{
   int min=arr[0];
   int max=arr[0];
   int index1=0,index2=0;
   for(int i=0;i<n;i++)
   {
      if(arr[i]>max)
      {
         max=arr[i];
         index1=i;
      }
      if(arr[i]<min)
      {
         min=arr[i];
         index2=i;
      }
   }
   return index1 - index2;
}
int main()
```

```
{
  int n,i,flag=0;
  printf("Enter n: ");
  scanf("%d",&n);
  int arr[n];
  for(i=0;i<n;i++)
  scanf("%d",&arr[i]);
 int difference = smallestlargest(arr,n);
 if(difference!=0)
 {
    printf("Difference is %d\n",difference);
 }
 else
 {
    printf("Invalid size\n");
 }
}
```

22. Palindrome:

```
#include <stdio.h>
#include <string.h>

int checkPalindrome(char str[], int i, int j)
{
   while(i < j)
   {
      if(str[i] != str[j])
         return 0;
      i++;
      j--;
```

```
    }
    return 1;
}

int canFormPalindrome(char str[])
{
    int i = 0, j = strlen(str) - 1;

    while(i < j)
    {
        if(str[i] != str[j])
        {
            // Try skipping one character once mismatch happens
            if(checkPalindrome(str, i+ 1, j))
                return 1;

            if(checkPalindrome(str, i, j- 1))
                return 1;

            return 0; // neither works
        }
        i++;
        j--;
    }

    return 1; // already palindrome
}

int main()
{
    char str[100];
```

```
    printf("Enter string: ");
    scanf("%s", str);

    if(canFormPalindrome(str))
        printf("Yes, it is palindrome\n");
    else
        printf("No, it is not a palindrome\n");

    return 0;
}
```

23. Replace Substring:

```
#include <stdio.h>
#include<string.h>
void replace_substring(char *str1, char*str2)
{
    char first[50],result[100];
    int i=0,j=0;


    while(str2[i]!='\0' && str2[i]!=' ')
    {
        first[j++]=str2[i++];
    }
    first[j]='\0';

    i=0;
    while(str1[i]!='\0' && str1[i]!=' ')
    {
        i++;
```

```
    }
    if(str1[i]==' ')
    i++;
    sprintf(result,"%s %s",first,str1+i);
    printf("Resulted string is %s\n",result);
}

int main()
{
 char str1[100];
 char str2[100];
 printf("Enter string : ");
 scanf("%[^\n]",str1);
  printf("Enter string : ");
 scanf(" %[^\n]",str2);
replace_substring(str1,str2);
    return 0;
}
```

Replace Occurences:

```
#include <stdio.h>
#include<string.h>
void replace_occurences(char*str)
{
    int i=0,j=0;
    char result[100];
    while(str[i]!='\0')
    {
        if(strncmp(&str[i],"bbc",3)==0)
        {
            result[j++]='$';
```

```
            i=i+3;
        }
        else if(strncmp(&str[i],"bbe",3)==0)
        {
            result[j++]='&';
            i=i+3;
        }
        else if(strncmp(&str[i],"bb",2)==0)
        {
            result[j++]='"';
            i=i+2;
        }
        else
        {
            result[j++]=str[i++];
        }
    }
    result[j]='\0';
    strcpy(str,result);
}
int main()
{
    char str[10];
    printf("Enter str1; ");
    scanf("%s",str);
    replace_occurences(str);
    printf("%s\n",str);
    return 0;
}
```

Duplicate Count:

```
#include <stdio.h>
#include<string.h>
void duplicate_count(int arr[],int n)
{
   int i,j,duplicate=0;
   for(i=0;i<n;i++)
   {
      int flag=0;
         for(j=0;j<i;j++)
         {
            if(arr[i]==arr[j])
            {
               flag=1;
               break;
            }
         }
         if(flag==0)
         {
            int count=0;
            for(int k=0;k<n;k++)
            {
               if(arr[i]==arr[k])
               {
                  count++;
               }
            }
            if(count>1)
            {
```

```
                duplicate++;
            }
        }
    }
    printf("Duplicate count is %d\n",duplicate);
}

int main()
{
    int n,i;
    printf("Enter n: ");
    scanf("%d",&n);
    int arr[n];
    for(i=0;i<n;i++)
    scanf("%d",&arr[i]);
    duplicate_count(arr,n);

    return 0;
}
```

Merge Array:

```
#include <stdio.h>
int main()
{
    int n1,n2,i,j;
    printf("Enter n1: ");
    scanf("%d",&n1);
    printf("Enter n2: ");
    scanf("%d",&n2);
    int arr[n1];
    int brr[n2];
```

```
    int total=n1+n2;
    int merge[total];
    for(i=0;i<n1;i++)
    scanf("%d",&arr[i]);
    for(i=0;i<n2;i++)
    scanf("%d",&brr[i]);

    for(i=0;i<total;i++)
    {
       if(i<n1)
       {
          merge[i]=arr[i];
       }
       else
       {
          merge[i]=brr[i-n1];
       }
    }

    for(i=0;i<total;i++)
    {
       printf("%d ",merge[i]);
    }
    return 0;
}
```

Column To Number:

```
#include <stdio.h>
#include<string.h>
int ColumnTitletoNumber(char*str)
{
   int i,result=0;
   for(i=0;str[i];i++)
   {
      result=result*26+(str[i]-'A'+1);
   }
   return result;
}
int main()
{
   char str[100];
   printf("Enter string: ");
   scanf("%s",str);
   int col_num=ColumnTitletoNumber(str);

  printf("Column corresponding to '%s' is %d\n",str,col_num);
   return 0;
}
```

Get Bits From Lsb With Position And N Bits:

```
#include <stdio.h>
int main()
{
  int num,n,pos;
  printf("Enter num: ");
  scanf("%d",&num);
  printf("Enter pos: ");
  scanf("%d",&pos);
  printf("Enter n: ");
  scanf("%d",&n);



  num=(num>>(pos-n+1))&((1<<n)-1);
  for(int i=n-1;i>=0;i--)
  {
     printf("%d",num>>i&1);
  }

   return 0;
}
```

Structure of car Brands and sorts using Mfg Year:

```
#include <stdio.h>

struct car
{
   char name[30];
   char brand[20];
   int year;
};

int main()
{
   int n,i=0,j=0;
printf("Enter how many entries: ");
scanf("%d",&n);
struct car cars[n];
for(i=0;i<n;i++)
{
   scanf("%s",cars[i].name);
   scanf("%s",cars[i].brand);
   scanf("%d",&cars[i].year);
}

for(i=0;i<n-1;i++)
{
   for(j=i+1;j<n;j++)
   {
      if(cars[i].year>cars[j].year)
      {
         struct car temp = cars[i];
```

```
            cars[i]=cars[j];
            cars[j]=temp;
        }
    }
}
printf("Sorted Cars:\n");
for(i=0;i<n;i++)
{
    printf("Name: %s, Brand: %s, Year: %d\n",
              cars[i].name, cars[i].brand, cars[i].year);
}
    return 0;
}
```

```
#include <stdio.h>

int main() {
    int a[5][5], target;

    // Input matrix
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    // Input target sum
    scanf("%d", &target);

    // Checking all directions
    for (int i = 0; i < 5; i++) {
        for (int j = 0; j < 5; j++) {
```

```
            // 1. RIGHT (j+1 within bounds)
            if (j + 1 < 5) {
                if (a[i][j] + a[i][j + 1] == target) {
                    printf("%d + %d: %d\n", a[i][j], a[i][j + 1], target);
                }
            }

            // 2. DOWN (i+1 within bounds)
            if (i + 1 < 5) {
                if (a[i][j] + a[i + 1][j] == target) {
                    printf("%d + %d: %d\n", a[i][j], a[i + 1][j], target);
                }
            }

            // 3. DIAGONAL DOWN-RIGHT
            if (i + 1 < 5 && j + 1 < 5) {
                if (a[i][j] + a[i + 1][j + 1] == target) {
                    printf("%d + %d: %d\n", a[i][j], a[i + 1][j + 1], target);
                }
            }
        }
    }

    return 0;
}
```

Second Largest:

```
#include <stdio.h>
int main()
{
int n,i;
printf("Enter n: ");
scanf("%d",&n);
int arr[n];
for(i=0;i<n;i++)
scanf("%d",&arr[i]);
int max=arr[0];
int min=arr[0];
for(i=0;i<n;i++)
{
   if(arr[i]>max)
   max=arr[i];
   if(arr[i]<min)
   min=arr[i];
}
int second=min;
for(i=0;i<n;i++)
{
   if(arr[i]<max && arr[i]>second)
   {
      second=arr[i];
   }
}
printf("Second largest is %d\n",second);
   return 0;
}
```

Smallest Positive Missing Number:

```
#include <stdio.h>
int main()
{
int n,i,j;
printf("Enter n: ");
scanf("%d",&n);
int arr[n];
for(i=0;i<n;i++)
scanf("%d",&arr[i]);
for(int missing=1;;missing++)
{
int found=0;
for(i=0;i<n;i++)
{
   if(arr[i]==missing)
   {
      found=1;
      break;
   }
}
if(!found)
{
   printf("The smallest positive number missed is: %d\n", missing);
   break;
}
}
   return 0;
}
```

Check Checker Board:

```
#include <stdio.h>
int isCheckerboard(int n, int matrix[n][n])
{
   int i,j;
   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         if(i+1<n)
         {
            if(matrix[i][j]==matrix[i+1][j])
             return 0;
         }
         if(j+1<n)
         {
            if(matrix[i][j]==matrix[i][j+1])
             return 0;
         }
      }
   }
   return 1;
}
int main()
{
int n,i,j;
printf("Enter n: ");
scanf("%d",&n);
int matrix[n][n];
for(i=0;i<n;i++)
```

```
{
   for(j=0;j<n;j++)
   {
      scanf("%d",&matrix[i][j]);
   }
}

if(isCheckerboard(n,matrix))
{
   printf("Valid checker board\n");
}
else
{
   printf("inValid checker board\n");
}
   return 0;
}
```

Row Sum & Column Sums:

```
#include <stdio.h>
int main()
{
int r1,c1,i,j;
printf("Enter r1 and c1: ");
scanf("%d%d",&r1,&c1);
int matrix[r1][c1];
for(i=0;i<r1;i++)
{
   for(j=0;j<c1;j++)
   {
```

```
        scanf("%d",&matrix[i][j]);
    }
}
 printf("Row sums: ");
for(i=0;i<r1;i++)
{
    int sum=0;
    for(j=0;j<c1;j++)
    {
       sum=sum+matrix[i][j];
    }
   printf("%d ",sum);
}
printf("\n");
printf("column sums: ");
for(i=0;i<c1;i++)
{
    int sum=0;
    for(j=0;j<r1;j++)
    {
       sum=sum+matrix[j][i];
    }
   printf("%d ",sum);
}
    return 0;
}
```

Remove Duplicates:

```
#include <stdio.h>
int main()
{
int n,i,j,flag;

printf("Enter n: ");
scanf("%d",&n);
int arr[n];
for(i=0;i<n;i++)
scanf("%d",&arr[i]);

for(i=0;i<n;i++)
{
   flag=0;
   for(j=0;j<i;j++)
   {
      if(arr[i]==arr[j])
      {
         flag=1;
         break;
      }
   }
   if(flag==0)
   {
      printf("%d ",arr[i]);
   }
}
   return 0;
}
```

WaveForm Pattern Matrix:

```
#include <stdio.h>
int main()
{
int r1,c1,i,j;
printf("Enter n: ");
scanf("%d%d",&r1,&c1);
int arr[r1][c1];
for(i=0;i<r1;i++)
{
   for(j=0;j<c1;j++)
   {
      scanf("%d",&arr[i][j]);
   }
}
 for(j=0;j<c1;j++)
 {
    if(j%2==0)
    {
       for(i=0;i<r1;i++)
       {
          printf("%d ",arr[i][j]);
       }
    }
    else
    {
       for(i=r1-1;i>=0;i--)
       {
          printf("%d ",arr[i][j]);
       }
```

```
    }
 }
   return 0;
}
```

Cows & Bulls:

```
#include <stdio.h>
int main()
{
char secret[100];
char guess[100];
printf("Enter secret msg: ");
scanf("%s",secret);
printf("Enter guess msg: ");
scanf("%s",guess);

int cows=0,bulls=0,i,j;

for(i=0;secret[i]!='\0';i++)
{
   if(secret[i]==guess[i])
   bulls++;
}
for(i=0;secret[i];i++)
{
   for(j=0;guess[j];j++)
   {
      if(i!=j && secret[i]==guess[j])
      {
         cows++;
```

```
            break;
        }
    }
}
printf("%dA%dB",bulls,cows);
    return 0;
}
```

Count Digits:

```
#include <stdio.h>
int count_dig(int num)
{
    if (num == 0)
        return 1;       // 0 has 1 digit

    if (num < 0)
        num = -num;     // handle negative numbers

    if (num < 10)
        return 1;
    return 1 + count_dig(num / 10);
}
int main()
{
    int num;
    printf("Enter num: ");
    scanf("%d", &num);
    int res = count_dig(num);
    printf("Digit count is %d\n", res);
    return 0;
}
```